Project Initialization and Planning Phase

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

PocketSmart AI is a GenAI-powered, cross-platform budget and recommendation assistant. It converts a user's budget and preferences into a structured, shop-able plan for home interiors, parties and jewelry.

Project Overview

Objective	Help users plan purchases for home decor, parties and jewelry within a fixed budget, using Gemini to allocate money and recommend products across Indian platforms
Scope	Three planners (Home, Party, Jewelry), user accounts and sessions, recommendation history, shopping deep links, offline fallback; no payments or live price scraping

Problem Statement

Description	Users overspend or waste hours because budgets are not allocated across categories and product data is scattered across many platforms
Impact	Solving this saves time and money, reduces decision stress and makes planned spending transparent

Proposed Solution

Approach	A FastAPI backend builds a structured prompt from validated inputs and calls Gemini (text, or text plus outfit image). The reply is parsed, sanitised and re-checked: totals are recomputed and items are trimmed so the plan never exceeds the budget. Search links are generated server-side
Key Features	- Home, Party and Jewelry planners - Multimodal outfit analysis for jewelry - Budget adherence check and calculation table - Shopping links per item (Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, ...) - Register/login, JWT sessions, history and dashboard - Offline fallback plan and Gemini model fallback chain

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Any laptop, 2 CPU cores, no GPU (inference runs in Google's cloud)
Memory	RAM specifications	4 GB minimum (server uses about 90 MB)
Storage	Disk space for data, models and logs	1 GB free; SQLite DB and uploaded images are small

Resource Type	Description	Specification / Allocation
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	google-genai, Pillow, bcrypt, python-jose, pydantic, python-dotenv, pytest
Development Environment	IDE	VS Code, Git, GitHub
Data		
Data	Source, size, format	No training dataset; user inputs at run time (JSON/form) and Gemini API responses (JSON); stored in SQLite